INCH-POUND

MIL-C-24640/3C
26 June 1995
SUPERSEDING
MIL-C-24640/3B
31 July 1989

MILITARY SPECIFICATION SHEET

* CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT,
FOUR CONDUCTOR, POWER, TYPE FX
(INCLUDING VARIATION TYPE FXA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

Construction
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33, and AWG size and dimensions in accordance with Table I.

SECOND - Component wires shall be MIL-W-81044/12 and shall be tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640. Identification codes applied using Method 1 or 3. Dimensions as shown in Table I.

THIRD - Four wires shall be cabled together.

FOURTH - Binder tape applied helically with overlap, at manufacturer's option.

CONSTRUCTION VARIANT FOR TYPE FX

FIFTH - Crosslinked polyolefin jacket. (See Table I for thickness).

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CONSTRUCTION VARIANT FOR TYPE FXA

FIFTH - Crosslinked polyolefin jacket. (See Table II for thickness).

SIXTH - Braided metal armor. Dimensions in accordance with Table II.

* TABLE I. Construction details - unarmored.

Pin M24640/3	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN	FX-3	16	0.056	0.069	0.075	0.031	0.262	0.282	63.0	4.98
-02UN	FX-4	14	.071	.085	.093	.034	.311	.335	92.6	3.18

* TABLE II. Construction details - armored.

Pin M24640/3	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01AN	FXA-3	16	0.056	0.069	0.075	0.031	0.312	0.332	80.0	4.98
-02AN	FXA-4	14	.071	.085	.093	.034	.361	.385	112.6	3.18

REQUIREMENTS:

Qualification required.
Construction details: See tables I and II.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Color: In accordance with MIL-STD-104, class 1; white preferred
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Component wire: 8.0 kilovolts (peak), or 3 Khz spark test, 5.5 KV, 100 percent test.

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Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F). Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	See tables I and II
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	Not applicable
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	5000
* Conductor continuity	No failure
Jacket flaws	Not applicable
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
* Cross link proof test (percent maximum)	
Jacket	50
Drip	Not applicable
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable

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	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged minimum	50
*	Elongation - percent of unaged, minimum	50
*	Permanence of printing (conductor) - cycles, method 1 only, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - (type FXA only) - conformance to material, construction and coverage	No failure
	Shield - conformance to material, construction and coverage	Not applicable
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	Not applicable
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
	Surface transfer impedance	
	Milliohms per meter, maximum	Not applicable
	EMP response, dB, minimum	Not applicable
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D:	
	Flame propagation	No failure

QUALIFICATION INSPECTION:

*	Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
	Aging and compatibility cable	No failure
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged minimum	50
*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
*	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5

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	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
*	Component wire	No failure

Unit ordering length: 500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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